

Stroke Prognosis & Clinical Scores

Clinical prediction scales for ischemic and hemorrhagic stroke outcomes.

STROKE PROGNOSIS SCALES

Ischemic Stroke: ASTRAL & PLAN

Hemorrhagic Stroke: ICH Score

ASTRAL Score (Acute Ischemic Stroke)

Predicts 90-day poor functional outcome (mRS > 2)

Predictor Variable	Points
Age	1 pt per 5 years
Severity (NIHSS)	1 pt per NIHSS pt
Time to admission >3h	2 pts
Range of visual fields (defect)	2 pts
Acute glucose (<3.7 or >7.3 mmol/L)	1 pt
Level of consciousness (impaired)	3 pts

Score vs. 90-Day Unfavorable Outcome (mRS > 2) Risk:

<20	23	31	35	40+
<5%	15%	50%	70%	>90%

PLAN Score (Acute Ischemic Stroke)

Bedside prediction of 30-day mortality/dependence

Predictor Domain & Variables	Points
Preadmission dependence / Cancer	1.5 pts each
Preadmission CHF / Atrial Fibrillation	1.0 pt each
Level of consciousness (reduced)	5.0 pts
Age (decades)	1 pt per decade (max 10)
Neurologic: Leg / Arm weakness	2 pts each
Neurologic: Aphasia or Neglect	1.0 pt

Score vs. 30-Day Mortality Risk:

<6	9-10	13	16	>19
0.7%	4.4%	15%	35%	>65%

ICH Score (Intracerebral Hemorrhage)

Predicts 30-day mortality in spontaneous ICH

Predictor Component	Points
GCS: 3-4 (2) 5-12 (1) 13-15 (0)	0-2 pts
Age: ≥ 80 years	1 pt
ICH Volume: ≥ 30 mL	1 pt
Intraventricular Hemorrhage (IVH): Present	1 pt
Infratentorial Origin of Hemorrhage	1 pt

Score vs. 30-Day Mortality Risk:

0	1	2	3	4	5-6
0%	13%	26%	72%	94%	100%

Modified Rankin Scale (mRS)

Standardized scale for post-stroke global disability

Grade Clinical Recovery State

- 0 No symptoms at all.
- 1 No significant disability despite symptoms.
- 2 Slight disability; independent in daily affairs.
- 3 Moderate disability; walks unassisted.
- 4 Moderately severe; unable to walk/body-care unassisted.
- 5 Severe disability; bedridden, constant nursing care.
- 6 Dead.

Prognostication Principles & Limitations

- ****Not for Care Limitations****: These clinical scores serve to quantify severity, improve inter-provider communication, and assist in counseling. They ****MUST NOT**** be used in isolation as the sole basis for withholding reperfusion therapies, surgical decompression, or withdrawing life-sustaining treatment (avoiding the self-fulfilling prophecy of poor outcome).
- ****Dynamic Evaluation****: Clinical trajectory over the first 24-72 hours is often more predictive of final recovery than any single point-in-time calculation upon hospital admission.

ASTRAL Score: Ntaios G, et al. *Stroke*. 2012;43:2170-6. PMID: 22738924

PLAN Score: O'Donnell MJ, et al. *Arch Intern Med*. 2012;172:1548-56. PMID: 23090225

ICH Score: Hemphill JC 3rd, et al. *Stroke*. 2001;32:891-7. PMID: 11283388

mRS Scale: van Swieten JC, et al. *Stroke*. 1988;19:604-7. PMID: 3363593